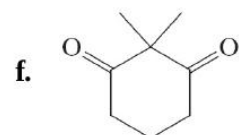
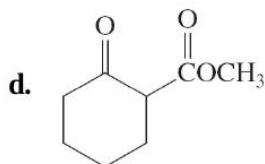
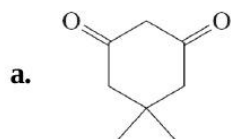
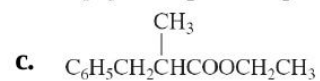
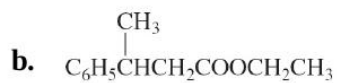


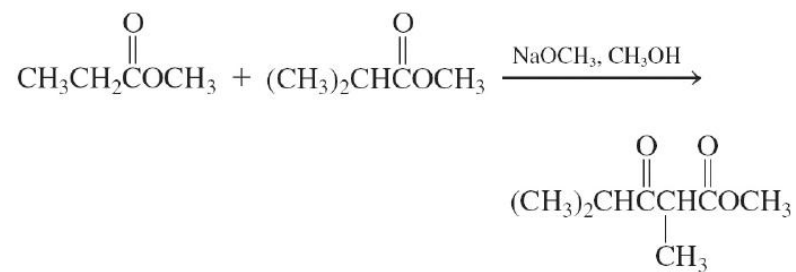
27. Arrange the following substances in order of increasing acidity.
 Estimate $\text{p}K_a$ values for each.



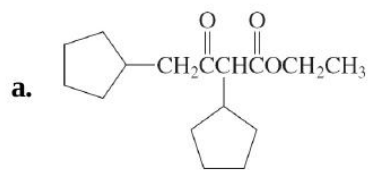
28. Give the expected results of the reaction of each of the following molecules (or combinations of molecules) with excess $\text{NaOCH}_2\text{CH}_3$ in $\text{CH}_3\text{CH}_2\text{OH}$, $\text{CH}_3\text{CH}_2\text{OH}$, followed by aqueous acidic work-up.



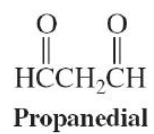
29. The following mixed Claisen condensation works best when one of the starting materials is present in large excess. Which of the two starting materials should be present in excess? Why? What side reaction will compete if the reagents are used in comparable amounts?



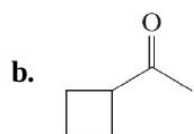
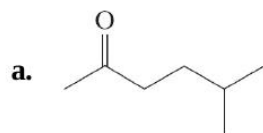
30. Suggest a synthesis of each of the following $\beta\beta$ -dicarbonyl compounds by Claisen or Dieckmann condensations.



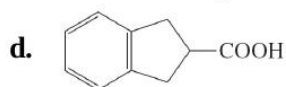
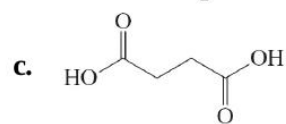
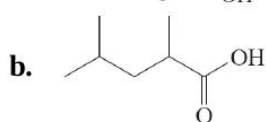
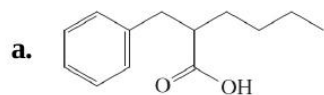
31. Do you think that propanedial, shown below, can be easily prepared by a simple Claisen condensation? Why or why not?



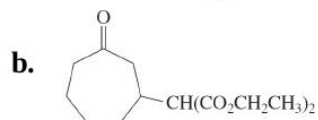
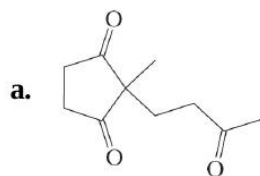
32. Devise a preparation of each of the following ketones by using the acetoacetic ester synthesis.



33. Devise a synthesis for each of the following four structures by using the malonic ester synthesis.



34. Use the methods described in [Section 23-3](#), with other reactions if necessary, to synthesize each of the following targets. In each case, your starting materials should include one aldehyde or ketone and one β -dicarbonyl compound.

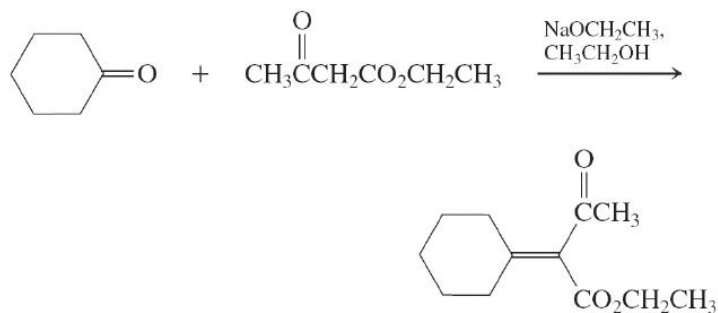


37. Write out, in full detail, the mechanism of the Michael addition of malonic ester to 3-buten-2-one in the presence of ethoxide ion. Be sure to indicate all steps that are reversible. Does the overall reaction appear to be exo- or endothermic? Explain why only a catalytic amount of base is necessary.

- 40.** Using the methods described in this chapter, design a multistep synthesis of each of the following molecules, making use of the indicated building

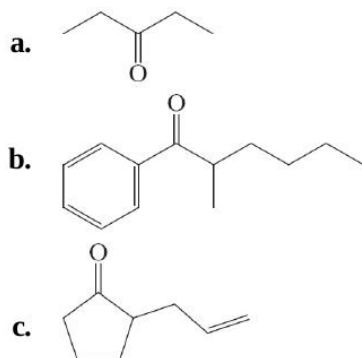
- 44.** Write chemical equations to illustrate all primary reaction steps that can occur between a base, such as ethoxide ion, and a carbonyl compound, such as acetaldehyde. Explain why the carbonyl carbon is not deprotonated to any appreciable extent in this system.

46. β -Dicarbonyl compounds condense with aldehydes and ketones that do not undergo self-aldol reaction. The products are α,β -unsaturated dicarbonyl derivatives, and the process goes by the colorful name of Knoevenagel condensation. **(a)** An example of a Knoevenagel condensation is given below. Propose a mechanism.



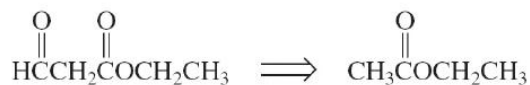
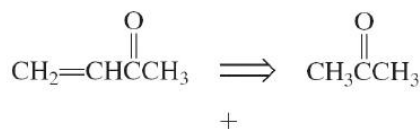
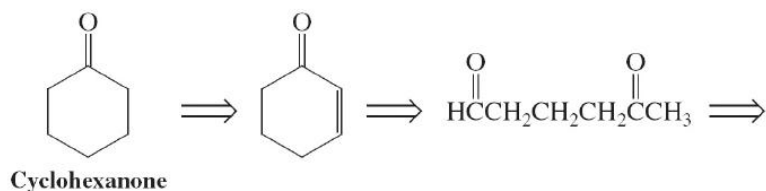
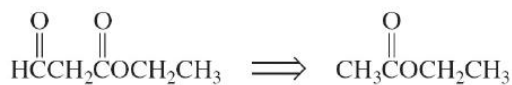
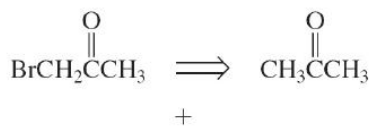
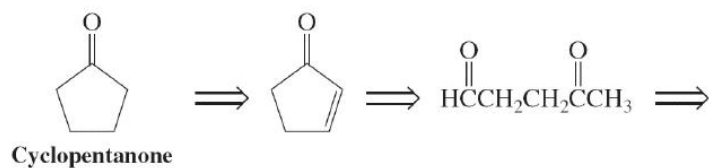
47. The following ketones cannot be synthesized by the acetoacetic ester method (why?), but they can be prepared by a modified version of it. The modification involves the preparation (by Claisen condensation)

and use of an appropriate 3-ketoester, $\text{RC}(=\text{O})\text{CH}_2\text{C}(=\text{O})\text{CH}_2\text{CH}_3$, containing an R group that appears in the final product. Synthesize each of the following ketones. For each, show the structure and synthesis of the necessary 3-ketoester as well.

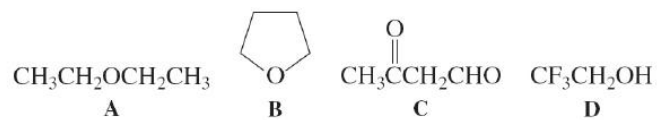


(Hint: Use a Dieckmann condensation.)

48. Some of the most important building blocks for synthesis are very simple molecules. Although cyclopentanone and cyclohexanone are readily available commercially, an understanding of how they can be made from simpler molecules is instructive. The following are possible retrosynthetic analyses ([Section 8-8](#)) for both of these ketones. Using them as a guide, write out a synthesis of each ketone from the indicated starting materials.

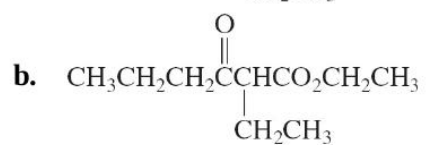
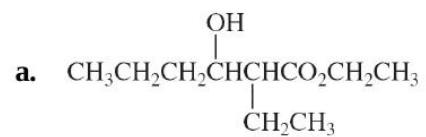


53. Two of the following four substances are more acidic than CH_3OH (i.e., two of these have K_a greater than methanol). Which ones?



- a. A and B;
- b. B and C;
- c. C and D;
- d. D and A;
- e. D and B.

54. The reaction of ethyl butanoate with sodium ethoxide in $\text{CH}_3\text{CH}_2\text{OH}$ gives



55. When acid A (below) is heated to 230°C, CO_2 and H_2O are evolved and a new compound is formed. Which one?

